Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/17/2007
PI Name:	Rittmann, Bruce Ph.D.		
Project Title:	A Novel Membrane Microbial Fuel Cell for Energy and Resource Recovery from Solid Organic Wastes		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5701	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	07/31/2004	**End Date:**	08/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 08/31/2006 per PI (12/06) Project moved from Northwestern U in January 2005, per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lueptow, Richard (Northwestern University)		
Grant/Contract No.:	NNJ04HG05G		
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Task Description:	The Membrane Microbial Fuel Cell (MMFC) is an innovative approach that could produce modest amounts of electricity while removing contaminants from the water and breaking down organic waste into water and other harmless compounds. The MMFC marries membrane technology with biotechnology. Bacteria catalytically oxidize the organic material in the MMFC to produce electrons that flow in an external circuit. Unlike a standard fuel cell, which operates at high temperature, the microbially catalyzed reactions in an MMFC occur at room temperature. Thus, the MMFC processes organic waste and wastewater in a way that helps produce pure water, generates electricity, and avoids extreme conditions in the spacecraft. Although the MMFC is a very new approach, the potential advantages are enormous: processing organic waste to produce pure water and electricity with small size and no extreme conditions. Thus, the MMFC addresses several aspects of wastewater and solid waste in the AHST Critical Path Roadmap. To determine if and how this technology can be brought into practice, we established the following objectives for the proposed study: a. Develop a bench-scale MMFC for laboratory testing b. Testing the MMFC with simple substrates to begin assessing its performance c. Establish the ecology and electrochemistry principles underlying the performance of an MMFC
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although the MMFC is still at the proof-of-concept state, its potential impacts could transform the way that society view materials now considered wastes. The MMFC will process biomass to produce pure water and electricity with high efficiency and without any air-pollution.
Task Progress:	The engineering of the MFC is at the level of proof of concept. During our Phase I Pilot Study, we accomplished the following tasks: 1. We created a simple MMFC by using a hollow-fiber membrane cathode (HFMC), in which the MFC cathode is attached directly onto a hollow-fiber membrane that supplied O2 for the cathode reaction. 2. We preformed kinetic characterization in single- and double-compartment MFC 3. We compared the rate of degradation of common fermentation byproducts (acetate, butyrate, propionate, ethanol, and H2) in a dual-compartment MFC. We made an experimental MMFC in which the cathode and anode are immersed in the liquid media without separating them with a membrane. This MMFC generated electricity successfully (188 mA) and maintained the current for weeks. Our design is unique and potential advantageous, because the MMFC design allows MFC electrodes to be placed directly into wastewater without the need of specialized compartments, simplifying the overall design. Our simple MMFC design also demonstrates the concept we proposed for integrating the membrane and electrode in an MFC. We tested the MFC with acetate, which is one of the simplest organic contaminants in ALS wastewater and a simple substrate that MFC bacteria can use. Our fundamental studies make it possible for us to identify critical operational parameters. Although initial start up of an MFC took about 60 days with anaerobic digester sludge as inoculum, once suitable bacteria accumulated, starting up a new electrode was a rapid process that occurred over only about a week. We must understand better what bacterial community was established on the electrode and how we can rapidly accumulate suitable bacterial community for MFCs; these are critical next steps. We assessed the dual-substrate limitation effects on the bacterial kinetics. The current output of MFC depended strongly on the working electrode potential, especially at potentials lower than -0.1 mV against the Ag/AgCl electrode. When the current output was plotted against the potential, the current showed the classic Monod like saturation response. Substrate concentration and the electrode potential controlled the output current. A high substrate concentration produces more current; however, one goal of MFC is to remove the substrate. Thus, our result illustrate one of the trade offs in MFC. The Monod-Tessier model appears to capture overall behavior of the current generation better than the Monod-Monod model. We plan to conduct more comprehensive studies on how the substrate concentration, biofilm density and structure, and the electrode potential together limit the current output response, because these are the factors that can be controlled via MFC design and operator. We tested the ability of MFC to remove typical fermentation byproducts acetate, propionate, butyrate, and H2. The fraction of electron of the substrates consumed (i.e. Coulombic efficiency) increased with the acclimation of bacteria. While acetate and ethanol were consumed fairly rapidly, butyrate and H2 consumption was slow. The presence of butyrate and H2 increased the current production in respect to the decay; however, their contribution to the overall current was small. The current production of propionate was much more favorable compared to butyrate, and further improvement in the current production was observed for ethanol and acetate. While the saturation type response was observed for propionate, saturation was not observed for acetate, indicating that the bacteria for the acetate biofilm were limited more by the anode potential than by the donor. The results indicate that fermentation, which converts complex mixtures of organic materials to simpler byproducts, such as ethanol and acetate, is essential for good MFC performance. MFC can generate electricity from many of simple alcohols, acids, and amino acid. More complex organic materials such as glucuronate can be fermented to ethanol, acetate, and lactic acids. A next crucial step will be identify conditions required to directly utilize or ferment other major ALS wastewater constituents, such as Igepon and creatine, in an MFC. The pH of the reactor was crucial operational variable for system performance. In general higher pH improved the system performance, and the pH dependencies were different for types of substrate utilized. While acetate had a most stringent pH requirement for maximum performance (pH > 7.3), propionate and ethanol was able to tolerate pH below 7 without significant loss in the current density. Wastewater that has a high pH value, such as those in Early Planetary Base Wastewater Ersatz, should be favorable for MFC performance.
Bibliography Type:	Description: (Last Updated: 06/15/2020)

Abstracts for Journals and Proceedings	Kato-Marcus A, Torres CI, Krajmalnik-Brown R, Rittmann BE. "Anode kinetics and microbial community for acetate and hydrogen consumption in Microbial Fuel Cell." International Society of Microbial Ecology, Vienna, Austria, August 2006. International Society of Microbial Ecology, August 2006. , Aug-2006